Frame dragging and spin-dependent transport in self-gravitating scalar rings

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Rotating scalar matter provides a setting in which local frame dragging and particle transport can be studied separately from the total angular momentum. We construct stationary, self-gravitating rings of independent canonical complex scalar fields and test them against field equations, global charges and resolution changes. Compact single-ring solutions contain ergoregions. Concentric counter-rotating rings retain local frame dragging at zero total angular momentum and admit a channel in which both matter densities are small relative to their respective peaks. Spinning test particles traverse this channel at matched conserved energy. In a separate single-ring experiment, reversing the particle spin changes arrival at a finite target into a first turnaround. Short Cartesian spacetime evolutions of a weaker ring show constraint errors approaching fourth order and convergent charge changes. The results concern stationary geometry, finite-distance test-particle transport and short-time numerical convergence.

I. INTRODUCTION

Rotating boson stars are self-gravitating complex scalar configurations with stationary stress tensors [1]. Their matter can have a toroidal distribution, and sufficiently compact solutions can contain ergoregions, where an observer cannot remain at fixed spatial coordinates [2]. These systems allow us to distinguish the total angular momentum measured far away from the local rotation of inertial frames.

We study that distinction in single rings and in concentric rings made of independent fields with opposite angular momenta. We then follow spinning test particles in the resulting geometries. Equal conserved launch energies allow the effect of spin orientation to be separated from a change in initial energy. A complementary evolution study tests whether a stationary ring can be interpolated onto a Cartesian grid with convergent evolution errors.

Section II defines the matter and particle models. Sections III and IV describe the stationary solutions and numerical setup. Section V defines the measurements and criteria used to assess them. Sections VI–IX give the principal results, and Sec. X draws their physical conclusions. The appendices collect notation and independent checks.

II. MODEL

A. Einstein gravity and independent scalar fields

We use geometric units, $G = c = 1$, metric signature $(-+++)$, and scalar mass m to set the length and time units. Each complex field χ_a gravitates through its own stress tensor; the fields have no direct coupling. For the mass-only configurations considered here the action is

$$S = \frac{1}{16\pi}\int d^4x\sqrt{-g}\left[R - 2\sum_a \mathcal{L}_a\right], \qquad \mathcal{L}_a = \nabla_\mu\chi_a\nabla^\mu\chi_a^* + m^2|\chi_a|^2, \tag{1}$$

where g is the determinant of the spacetime metric $g_{\mu\nu}$, R its Ricci scalar, ∇_μ the metric covariant derivative and the star denotes complex conjugation. The index a labels independent fields, whose scalar Lagrangian is $\mathcal{L}_a$. Variation gives

$$R_{\mu\nu} = 2\sum_a \mathrm{Re}(\partial_\mu\chi_a\partial_\nu\chi_a^*) + m^2\sum_a |\chi_a|^2 g_{\mu\nu}, \tag{2}$$

$$(\Box - m^2)\chi_a = 0, \tag{3}$$

where $R_{\mu\nu}$ is the Ricci tensor and $\Box = \nabla_\mu\nabla^\mu$. The canonical kinetic sign gives a nonnegative contraction of the stress tensor with any null vector. Each configuration has a regular centre and one asymptotically flat end.

B. Stationary geometry and scalar phase

The stationary metric is written in quasi-isotropic spherical coordinates (t, r, θ, φ) as

$$ds^2 = -e^f dt^2 + B^2 e^{-f}\left[e^\nu(dr^2 + r^2 d\theta^2) + r^2\sin^2\theta(d\varphi - w dt)^2\right], \tag{4}$$

where f, B, ν, w depend on r and θ. The function w describes angular frame dragging. The scalar phase is

$$\chi_a = A_a(r,\theta)e^{i\omega_a t + ik_a\varphi}, \tag{5}$$

where A_a is a real amplitude, ω_a its frequency and k_a its integer azimuthal winding. Independent fields add stress tensors without interference between their phases. The cylindrical coordinates are $\rho = r\sin\theta$ and $z = r\cos\theta$; plotted radii are coordinate distances.

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C. Spinning test particles

We treat the particles in the pole–dipole approximation, retaining momentum and spin but neglecting their backreaction on the scalar spacetime. The Mathisson–Papapetrou–Dixon equations [3] are

$$\frac{Dp^\mu}{d\tau} = -\frac{1}{2} R^\mu{}_{\nu\alpha\beta} u^\nu S^{\alpha\beta}, \tag{6}$$

$$\frac{DS^{\mu\nu}}{d\tau} = p^\mu u^\nu - p^\nu u^\mu, \qquad p_\mu S^{\mu\nu} = 0, \tag{7}$$

where τ is proper time, u^μ the tangent velocity, p^μ momentum, $S^{\mu\nu}$ the spin tensor and $R^\mu{}_{\nu\alpha\beta}$ the Riemann tensor. The final relation is the Tulczyjew–Dixon spin supplementary condition. We evolve the full momentum–velocity relation and monitor its timelike normalization, particle mass and spin magnitude.

III. STATIONARY SOLUTIONS AND INITIAL DATA

A. Construction and independent validation

Figure 1 separates the stationary solve from its independent checks and the subsequent spacetime evolution. A host process selects the graphics processing unit (GPU) and enables double precision before JAX compiles the field residual. The compiled residual and its automatically differentiated Jacobian execute on the GPU. Jacobian columns are evaluated in blocks to limit temporary memory, while the assembled dense matrix remains on the device for LU factorization, the Newton correction and line-search residual evaluations. Here LU denotes triangular matrix factorization.

A tensor-product Chebyshev representation covers a compactified radial coordinate and the polar angle. Analytic factors remove the scalar's known axis and parity behavior before Newton iteration solves for the remaining smooth functions and frequencies. Asymptotic flatness fixes the time normalization. For the counter-rotating pair, one scalar amplitude is adjusted to make the total angular momentum vanish.

Every reported stationary model passes three independently recovered resolutions. The checks include Einstein equations not used in the solve, the Klein–Gordon equations evaluated in both intrinsic and Cartesian coordinates, boundary conditions, spectral tails and independently evaluated volume and surface charges. A small Newton residual alone is insufficient: the equations not used by the solver must also hold.

B. Cartesian initial data

For evolution we evaluate the stationary metric and both real scalar components at the Cartesian cell centres. The lapse α, shift β^i and spatial metric γ_{ij} are defined by

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \tag{8}$$

with $\alpha = e^{f/2}$ and $\beta^\varphi = -w$ for Eq. (4). Spatial indices are i, j. Metric derivatives determine the extrinsic curvature before sampling. The complex evolution field is $\Phi = \chi/\sqrt{4\pi}$, with normal momentum $\Pi = n^\mu \partial_\mu \Phi$, where n^μ is the future unit normal to a constant-time slice. A local Cartesian polynomial representation supplies regular derivatives at the centre and is checked against the field equations and curvature before interpolation.

IV. NUMERICAL SETUP

Table I separates the physical models from the experiments performed on them. The stationary ergoregion examples, particle backgrounds and evolved ring are distinct configurations.

The Cartesian evolutions use GRTeclyn [4], the conformal covariant Z4 formulation (CCZ4), fourth-order spatial derivatives and fourth-order Runge–Kutta time integration. The initial stationary lapse and shift evolve with $1 + \log$ slicing and a Gamma-driver shift condition. The uniform cube has side length $mL = 32$, with time step $\Delta t/\Delta x = 0.02$ and Sommerfeld conditions on each outer face. Constraint norms use the same interior sphere, $mr < 8$, at every resolution. These choices define the short-time comparison.

Particle integrations use two Runge–Kutta tolerances, spin reversal and simultaneous background/spin reversal. An independent axial first-integral calculation checks the near-turning trajectories. Appendix B collects the principal numerical tolerances.

V. DIAGNOSTICS AND CRITERIA

A. Ergoregions and frame dragging

The stationary Killing vector is ∂_t. Its squared norm is

$$g_{tt} = -\alpha^2 + \gamma_{\varphi\varphi} w^2 = \alpha^2(\mathcal{D}^2 - 1), \qquad \mathcal{D} = \frac{\sqrt{\gamma_{ij}\beta^i\beta^j}}{\alpha}. \tag{9}$$

An ergoregion requires $g_{tt} > 0$, consistently resolved across the stationary grids. The dimensionless ratio $\mathcal{D}$ compares the shift norm with the lapse. It is a metric diagnostic; a local observer measures light speed c. A zero-angular-momentum observer (ZAMO) rotates at angular velocity w. A clock held at fixed coordinates instead has rate $d\tau/dt = \sqrt{-g_{tt}}$ wherever that observer is timelike.

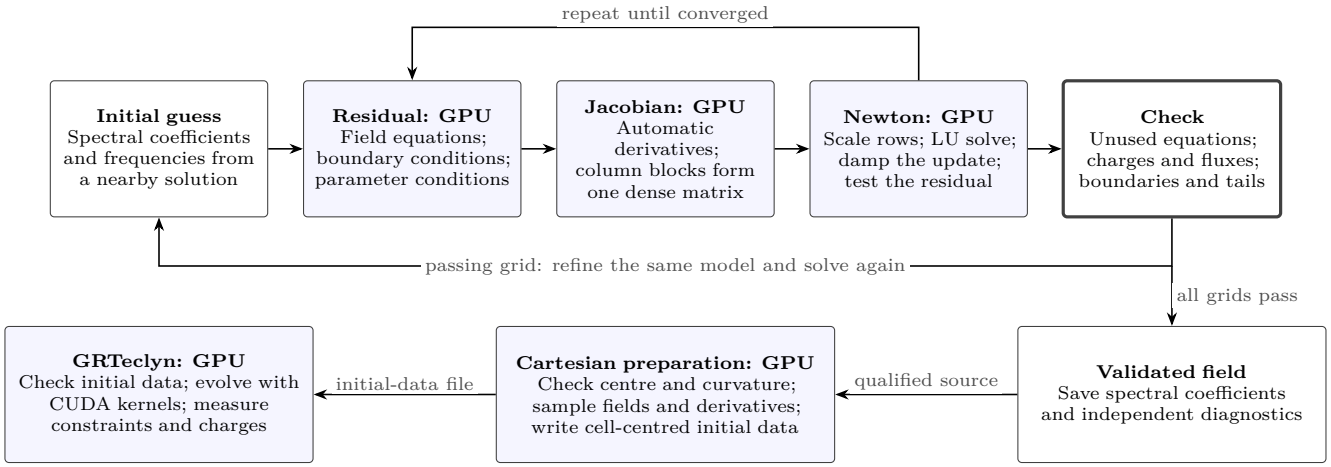


FIG. 1. Stationary solution and Cartesian evolution workflow. Thin boxes construct or evolve fields; the heavy box tests equations and charges independently of the solve. Blue boxes identify the main GPU calculations; the host controls iteration and file output. Only passing stationary and Cartesian-origin checks permit preparation of evolution data.

TABLE I. Setup of the principal experiments. Spectral grids list radial and angular degrees; Cartesian grids list cells per spatial direction. The mass m fixes units, and particle endpoints use the coordinate z.

Model	Matter parameters	Resolutions	Role
E	Single field, $k = +1$	$160 \times 20/192 \times 24/224 \times 28$	Stationary ergoregion
C	Independent fields, $k_1 = +1$, $k_2 = -12$	$192 \times 16/224 \times 18/256 \times 20$	Counter-rotation and channel
P	Single field, $\omega/m = 0.73$	$96 \times 12/128 \times 16/160 \times 20$	Axial spin selection
U	Single field, $\omega/m = 0.70$	$96^3/128^3/192^3$	Evolution through $mt = 2$

B. Angular momentum and the low-density channel

For the phase convention in Eq. (5), the scalar charges obey

$$Q_a = \frac{1}{4\pi}\int \frac{(\omega_a + k_a w)A_a^2}{\alpha}\, dV, \qquad J_a = -k_a Q_a, \qquad J = \sum_a J_a, \tag{10}$$

where dV is proper spatial volume, Q_a is the Noether charge and J_a the component angular momentum. Independent surface measurements test J. The asymptotic coefficient $w \sim 2J/r^3$ fixes only the leading exterior rotational term; cancelling it allows nonzero interior dragging.

Let $\epsilon_a = T^{(a)}_{\mu\nu} n^\mu n^\nu$ be the Eulerian energy density of component a, with stress tensor $T^{(a)}_{\mu\nu}$. Its relative density is

$$\eta_a(\rho, z) = \frac{\epsilon_a(\rho, z)}{\epsilon_{a,\mathrm{peak}}}, \qquad \eta_a < 10^{-6} \quad \text{for each component,} \tag{11}$$

where $\epsilon_{a,\mathrm{peak}}$ is that component's equatorial peak. The last outer crossing for the inner field and the first inner crossing for the outer field define the equatorial interval. We then test a common finite-height tube and the density along every particle trajectory. This definition retains the diffuse scalar tails.

C. Equal-energy particle comparisons

A Killing vector ξ^μ supplies a conserved particle quantity,

$$C_\xi = p_\mu \xi^\mu + \frac{1}{2} S^{\mu\nu} \nabla_\mu \xi_\nu, \qquad E = -C_{\partial_t}, \tag{12}$$

where E includes the spin contribution. All compared launches have equal E and unit particle rest mass. Passage means reaching the specified finite target before a first turnaround while preserving particle invariants and the timelike and supplementary conditions. Spin selection requires opposite orientations to give different endpoints under tolerance changes and independent timing checks.

VI. SINGLE RINGS WITH ERGOREGIONS

Compact single-ring solutions contain a resolved region with $g_{tt} > 0$. Figure 2(a,b) relates its position to the matter density; panels (c,d) show the lapse and the two observer clock rates. Panels (e,f) distinguish angular frame

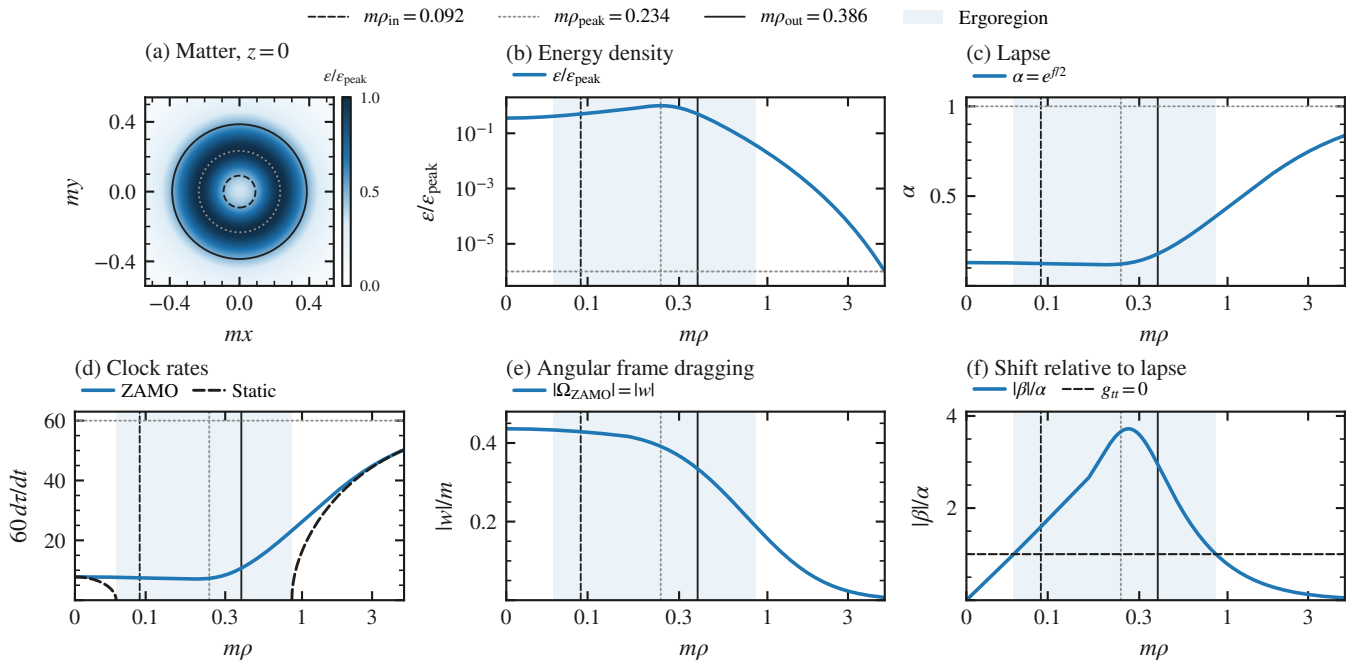


FIG. 2. Equatorial matter, density, lapse, clock rates, angular dragging and shift/lapse ratio for a compact single ring, against coordinate position or radius. Vertical lines mark the density peak and half-peak crossings; shading marks the ergoregion. The metric supports an ergoregion across the matter distribution and its surrounding tail.

dragging from $\mathcal{D}$. Static observers cease to be timelike inside the shaded region, while the ZAMO clock remains well defined.

The strongest independently verified example has $\mathcal{D}_{\rm max} \simeq 5.23$. This is the largest accepted value in the sampled stationary family. Table II gives a separate quantitative example for the model plotted in Fig. 2.

VII. COUNTER-ROTATING RINGS AND A LOW-DENSITY CHANNEL

Two independent fields with opposite winding form concentric rings whose component angular momenta cancel. The volume and infinity measurements of the total angular momentum agree within the independent numerical checks. Nevertheless, the angular frame dragging is nonzero in the interior.

Figure 3(a,b) shows the spatial separation and the component-normalized density profiles. Both satisfy Eq. (11) in the shaded interval, whose width is $m\Delta\rho \simeq 7.61$. Panels (c,d) show the lapse and clock rates. Panels (e,f) display the signed dragging and shift/lapse ratio. The cancellation of the global angular momentum therefore coexists with a resolved, locally rotating geometry.

All 8 equal-energy particle tracks reached the opposite finite-height endpoint while remaining inside the validated tube. The controls include zero spin, both axial spin orientations and simultaneous background/spin reversal at two tolerances. The maximum density encountered along the trajectories remained below the component-wise cutoff. The measured spin-dependent coordinate transit-time difference appears in Table II.

VIII. SPIN-DEPENDENT ARRIVAL AND TURNAROUND

The axial single-ring experiment tests whether spin orientation changes arrival at a prescribed target at fixed conserved energy. The ring's angular momentum is negative. A particle with positive axial spin reaches the target, whereas its negative-spin counterpart first turns just before it; the zero-spin control reaches the target. Figure 4(a) shows the full trajectories, and panel (b) resolves their separation near the endpoint. The target and spin magnitude are specified in Table II and the figure legend, where S_z denotes the signed axial particle spin in unit-rest-mass coordinates.

Independent first-integral quadratures reproduce the nonmarginal arrival times, and direct substitution checks the reduced dynamics against the full Cartesian particle equations. This establishes spin-dependent finite-target selection in the stationary background.

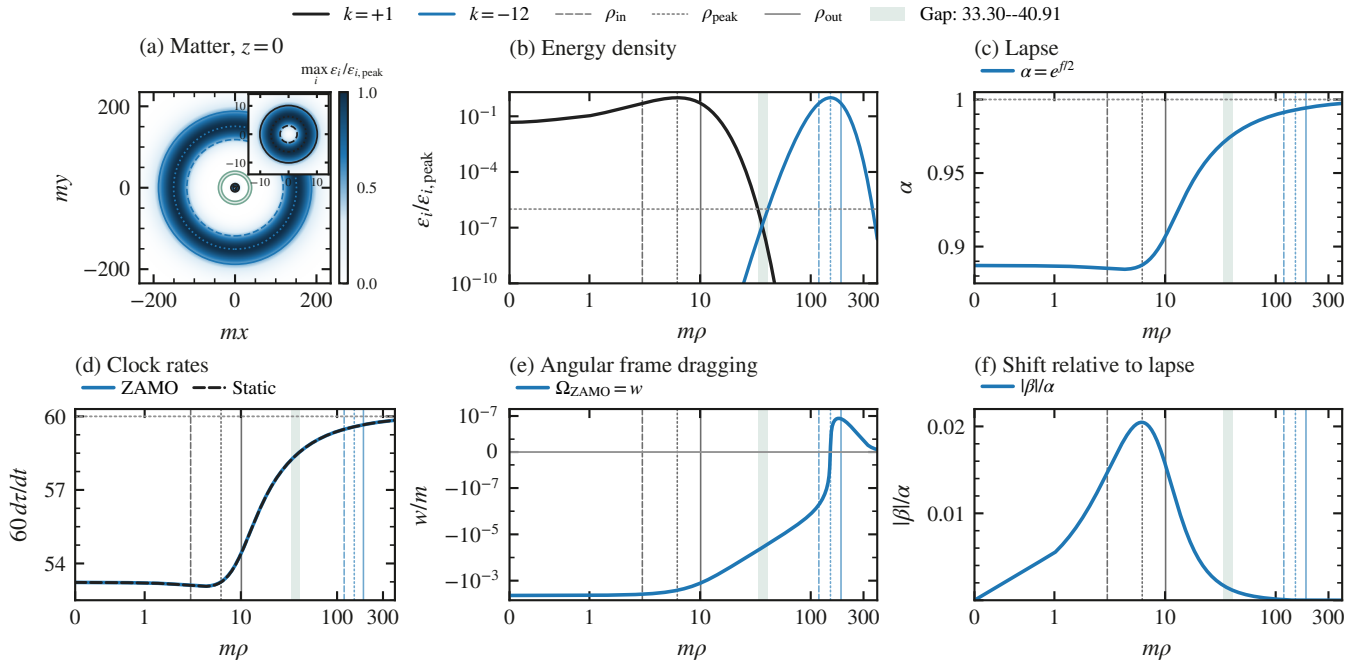


FIG. 3. Equatorial matter distribution, component densities, lapse, clock rates, signed angular dragging and shift/lapse ratio for the counter-rotating pair. Black and blue distinguish the scalar components; vertical lines mark half-peak crossings and peaks, and green marks the low-density interval. Zero total angular momentum retains local frame dragging.

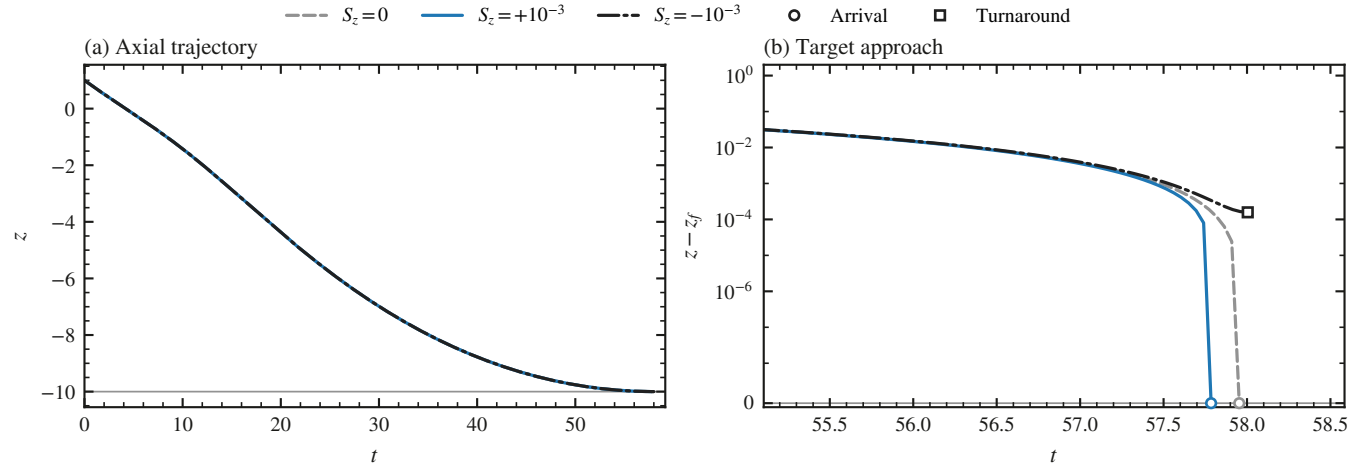


FIG. 4. Axial coordinate against time and distance from the target against time for equal-energy particles in the single-ring background. Grey, blue and black denote zero, positive and negative axial spin; circles mark arrivals and the square marks a first turnaround. Reversing the spin changes arrival into turnaround near the finite target.

IX. SHORT-TIME CARTESIAN EVOLUTION AND CONVERGENCE

The weaker single ring provides a short-time test of the Cartesian initial data and spacetime evolution. Three uniform grids evolved the same unperturbed stationary solution through $mt = 2$. Figure 5(a,b) shows decreasing interior constraint norms and second- and fourth-order rescalings. Panel (c) gives the orders measured between adjacent resolutions. Because the exact constraints vanish, their norms can be compared directly under resolution rescaling.

The finer-pair constraint orders approach fourth order (Table II). Charge changes also decrease with resolution, as shown in Fig. 5(d). Panels (e,f) compare the signed charge differences and the inferred orders for charge change and matter angular momentum. These are measured using the unequal grid spacings, rather than

TABLE II. Representative measured results. Lengths and times use the scalar mass unit; E is the spin-corrected energy of a unit-rest-mass particle. The dragging maximum and the plotted ergoregion example refer to distinct stationary configurations.

Experiment	Quantity	Measurement
Strongest sampled single ring	$\mathcal{D}_{\max}$	5.23
Plotted ergoregion ring	$\max g_{tt}$	0.2468
Counter-rotating pair	$J_1,\ J_2$	$-0.99196,\ +0.99196$
Counter-rotating pair	$m\rho$ interval	$[33.30, 40.91]$
Between-ring transport	E	1.125109
Between-ring transport	$mz_{\mathrm{launch}} \to mz_{\mathrm{target}}$	$+10 \to -10$
Between-ring transport	$m(t_+ - t_-)$	3.59×10^{-6}
Axial single-ring transport	E	0.88029376
Axial single-ring transport	mz_{turn} for negative spin	-9.99984248
Evolution at $mt = 2$	Hamiltonian order, finer pair	4.01
Evolution at $mt = 2$	Momentum order, finer pair	3.83
Evolution at $mt = 2$	Charge-change self-convergence order	4.68

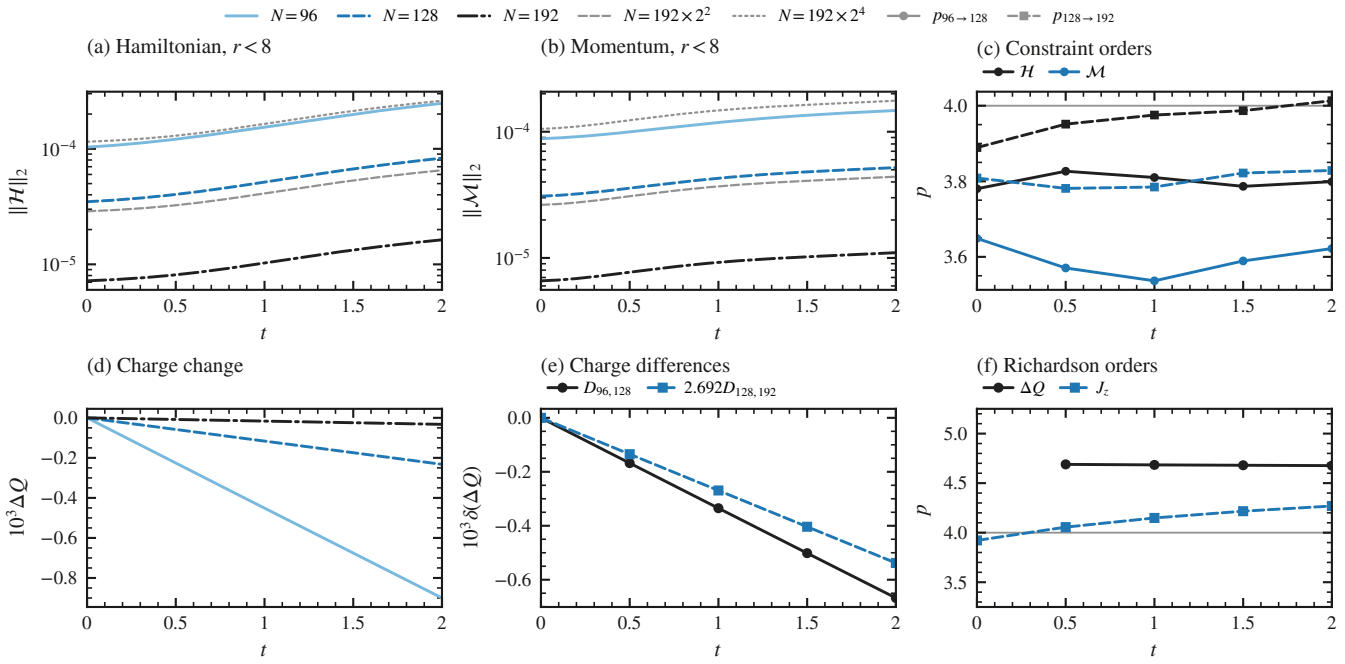


FIG. 5. Interior constraint norms, measured orders and charge changes against time for three Cartesian resolutions. The legend identifies resolutions, reference rescalings and adjacent-grid pairs; D denotes the charge-change difference in Eq. (B1). The constraint and observable errors decrease consistently under refinement.

assuming a fixed refinement ratio. Together these tests show short-time numerical convergence for the evolved configuration.

X. CONCLUSIONS

Self-gravitating canonical scalar rings provide stationary geometries with resolved ergoregions and spin-dependent test-particle transport. A pair of counter-rotating fields retains local frame dragging despite cancellation of its total angular momentum. Their separated density distributions leave a finite-height, low-density channel traversed by the tested particles. In the single-ring experiment, changing spin orientation at fixed conserved energy changes arrival at a finite target into turnaround. The weaker ring's Cartesian evolution exhibits short-time constraint and observable convergence. These findings connect stationary field structure to finite-distance transport within the specified matter and test-particle models.

ACKNOWLEDGMENTS

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Appendix A: Notation

Spatial norms use γ_{ij}. The principal notation is collected below.

Symbol	Meaning
$\chi_a, A_a, \omega_a, k_a$	Scalar field, amplitude, frequency, winding
m	Common scalar mass
f, B, ν, w	Stationary metric functions
ρ, z	Cylindrical coordinates
$\alpha, \beta^i, \gamma_{ij}$	Lapse, shift, spatial metric
$\mathcal{D}$	Shift norm divided by lapse
Q_a, J_a, J	Scalar charge, component and total spin
ϵ_a, η_a	Energy density and relative density
$p^\mu, u^\mu, S^{\mu\nu}$	Particle momentum, velocity, spin tensor
E, τ	Conserved particle energy, proper time
Φ, Π	Evolution field and normal momentum
h, p	Grid spacing and convergence order
$\Delta Q, D_{ab}$	Charge change and its grid difference

Appendix B: Independent validation and convergence estimates

The stationary equations and charge-flux comparisons use the tolerance 10^{-8}. Spectral tails, relative changes between stationary resolutions and the matter-density cutoff are tested separately. Particle mass, spin, conserved energy and supplementary-condition errors are required to remain below 10^{-9} in their stated normalization. The independent nonmarginal timing comparison uses 10^{-7}. These are acceptance tolerances for the numerical tests, not error bars on every derived physical observable.

For the Cartesian charge, let $Q_N(t)$ be the finite-box scalar charge at resolution N, and $\Delta Q_N(t) = Q_N(t) - Q_N(0)$. With $h_a > h_b > h_c$ the three grid spacings, define

$$D_{ab} = \Delta Q_a - \Delta Q_b, \qquad \frac{D_{ab}}{D_{bc}} = \frac{h_a^p - h_b^p}{h_b^p - h_c^p}. \tag{B1}$$

The order p is fitted from signed resolved differences. This assumes a leading power-law error term; the resulting extrapolation is an asymptotic estimate. The expected fourth-order difference factor for these grids is 2.692, the rescaling used in Fig. 5(e).

Appendix C: Numerical data and figure provenance

The accompanying repository stores stationary fields, portable evolution diagnostics and the numerical inputs for the figures. Each stationary and particle study retains its configuration, checks and checksums. The manuscript number ledger links quoted measurements to specific archived diagnostic entries; its checker verifies source hashes before generating the macros. Figure assets are read directly from the repository's results directory.

[1] B. Kleihaus, J. Kunz and M. List, "Rotating boson stars and Q-balls," Phys. Rev. D **72**, 064002 (2005), arXiv:gr-qc/0505143.

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